



National  
Qualifications  
2023

**X857/75/11**

**Physics  
Relationships sheet**

WEDNESDAY, 17 MAY

1:00 PM – 3:30 PM

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\* X 8 5 7 7 5 1 1 \*

$$d = vt$$

$$d = \bar{v}t$$

$$s = vt$$

$$s = \bar{v}t$$

$$a = \frac{v-u}{t}$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$Q = It$$

$$V = IR$$

$$V_2 = \left( \frac{R_2}{R_1 + R_2} \right) V_S$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$P = \frac{E}{t}$$

$$P = IV$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

$$E_h = cm\Delta T$$

$$E_h = ml$$

$$p = \frac{F}{A}$$

$$p_1 V_1 = p_2 V_2$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{pV}{T} = \text{constant}$$

$$f = \frac{N}{t}$$

$$v = f\lambda$$

$$T = \frac{1}{f}$$

$$A = \frac{N}{t}$$

$$D = \frac{E}{m}$$

$$H = Dw_r$$

$$\dot{H} = \frac{H}{t}$$

## Additional Relationships

### Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

### Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

### Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

## Electron arrangements of elements

Group 1      Group 2  
(1)

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| 1<br><b>H</b><br>1                        | 4<br><b>He</b><br>2                       |
| Hydrogen                                  | (2)                                       |
| 3<br><b>Li</b><br>2, 1                    | 4<br><b>Be</b><br>2, 2                    |
| Lithium                                   | Beryllium                                 |
| 11<br><b>Na</b><br>2, 8, 1                | 12<br><b>Mg</b><br>2, 8, 2                |
| Sodium                                    | Magnesium                                 |
| 19<br><b>K</b><br>2, 8, 8, 1              | 20<br><b>Ca</b><br>2, 8, 8, 2             |
| Potassium                                 | Calcium                                   |
| 37<br><b>Rb</b><br>2, 8, 18, 8, 1         | 38<br><b>Sr</b><br>2, 8, 18, 8, 2         |
| Rubidium                                  | Strontium                                 |
| 55<br><b>Cs</b><br>2, 8, 18, 18, 8, 1     | 56<br><b>Ba</b><br>2, 8, 18, 18, 8, 2     |
| Caesium                                   | Barium                                    |
| 87<br><b>Fr</b><br>2, 8, 18, 32, 18, 8, 1 | 88<br><b>Ra</b><br>2, 8, 18, 32, 18, 8, 2 |
| Francium                                  | Radium                                    |

### Key

|                      |
|----------------------|
| Atomic number        |
| Symbol               |
| Electron arrangement |
| Name                 |

### Transition elements

|                                           |                                         |                                         |                                         |                                         |                                          |                                         |                                         |                                         |                                         |
|-------------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| (3)                                       | (4)                                     | (5)                                     | (6)                                     | (7)                                     | (8)                                      | (9)                                     | (10)                                    | (11)                                    | (12)                                    |
| 21<br><b>Sc</b><br>2, 8, 9, 2             | 22<br><b>Ti</b><br>2, 8, 10, 2          | 23<br><b>V</b><br>2, 8, 11, 2           | 24<br><b>Cr</b><br>2, 8, 13, 1          | 25<br><b>Mn</b><br>2, 8, 13, 2          | 26<br><b>Fe</b><br>2, 8, 14, 2           | 27<br><b>Co</b><br>2, 8, 15, 2          | 28<br><b>Ni</b><br>2, 8, 16, 2          | 29<br><b>Cu</b><br>2, 8, 18, 1          | 30<br><b>Zn</b><br>2, 8, 18, 2          |
| Scandium                                  | Titanium                                | Vanadium                                | Chromium                                | Manganese                               | Iron                                     | Cobalt                                  | Nickel                                  | Copper                                  | Zinc                                    |
| 39<br><b>Y</b><br>2, 8, 18, 9, 2          | 40<br><b>Zr</b><br>2, 8, 18, 10, 2      | 41<br><b>Nb</b><br>2, 8, 18, 12, 1      | 42<br><b>Mo</b><br>2, 8, 18, 13, 1      | 43<br><b>Tc</b><br>2, 8, 18, 13, 2      | 44<br><b>Ru</b><br>2, 8, 18, 15, 1       | 45<br><b>Rh</b><br>2, 8, 18, 16, 1      | 46<br><b>Pd</b><br>2, 8, 18, 18, 0      | 47<br><b>Ag</b><br>2, 8, 18, 18, 1      | 48<br><b>Cd</b><br>2, 8, 18, 18, 2      |
| Yttrium                                   | Zirconium                               | Niobium                                 | Molybdenum                              | Technetium                              | Ruthenium                                | Rhodium                                 | Palladium                               | Silver                                  | Cadmium                                 |
| 57<br><b>La</b><br>2, 8, 18, 18, 9, 2     | 72<br><b>Hf</b><br>2, 8, 18, 32, 10, 2  | 73<br><b>Ta</b><br>2, 8, 18, 32, 11, 2  | 74<br><b>W</b><br>2, 8, 18, 32, 12, 2   | 75<br><b>Re</b><br>2, 8, 18, 32, 13, 2  | 76<br><b>Os</b><br>2, 8, 18, 32, 14, 2   | 77<br><b>Ir</b><br>2, 8, 18, 32, 15, 2  | 78<br><b>Pt</b><br>2, 8, 18, 32, 17, 1  | 79<br><b>Au</b><br>2, 8, 18, 32, 18, 1  | 80<br><b>Hg</b><br>2, 8, 18, 32, 18, 2  |
| Lanthanum                                 | Hafnium                                 | Tantalum                                | Tungsten                                | Rhenium                                 | Osmium                                   | Iridium                                 | Platinum                                | Gold                                    | Mercury                                 |
| 89<br><b>Ac</b><br>2, 8, 18, 32, 18, 9, 2 | 104<br><b>Rf</b><br>2, 8, 18, 32, 10, 2 | 105<br><b>Db</b><br>2, 8, 18, 32, 11, 2 | 106<br><b>Sg</b><br>2, 8, 18, 32, 12, 2 | 107<br><b>Bh</b><br>2, 8, 18, 32, 13, 2 | 108<br><b>Has</b><br>2, 8, 18, 32, 14, 2 | 109<br><b>Mt</b><br>2, 8, 18, 32, 15, 2 | 110<br><b>Ds</b><br>2, 8, 18, 32, 17, 1 | 111<br><b>Rg</b><br>2, 8, 18, 32, 18, 1 | 112<br><b>Cn</b><br>2, 8, 18, 32, 18, 2 |
| Actinium                                  | Rutherfordium                           | Dubnium                                 | Seaborgium                              | Bohrium                                 | Hassium                                  | Meitnerium                              | Darmstadtium                            | Roentgenium                             | Copernicium                             |

|                                        |                                        |                                        |                                        |                                        |                                        |
|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| 2<br><b>He</b><br>2                    | 6<br><b>C</b><br>2, 4                  | 7<br><b>N</b><br>2, 5                  | 8<br><b>O</b><br>2, 6                  | 9<br><b>F</b><br>2, 7                  | 10<br><b>Ne</b><br>2, 8                |
| Helium                                 | Carbon                                 | Nitrogen                               | Oxygen                                 | Fluorine                               | Neon                                   |
| (13)                                   | (14)                                   | (15)                                   | (16)                                   | (17)                                   | (18)                                   |
| 5<br><b>B</b><br>2, 3                  | 6<br><b>C</b><br>2, 4                  | 7<br><b>N</b><br>2, 5                  | 8<br><b>O</b><br>2, 6                  | 9<br><b>F</b><br>2, 7                  | 10<br><b>Ne</b><br>2, 8                |
| Boron                                  | Carbon                                 | Nitrogen                               | Oxygen                                 | Fluorine                               | Neon                                   |
| 13<br><b>Al</b><br>2, 8, 3             | 14<br><b>Si</b><br>2, 8, 4             | 15<br><b>P</b><br>2, 8, 5              | 16<br><b>S</b><br>2, 8, 6              | 17<br><b>Cl</b><br>2, 8, 7             | 18<br><b>Ar</b><br>2, 8, 8             |
| Aluminium                              | Silicon                                | Phosphorus                             | Sulfur                                 | Chlorine                               | Argon                                  |
| 31<br><b>Ga</b><br>2, 8, 18, 3         | 32<br><b>Ge</b><br>2, 8, 18, 4         | 33<br><b>As</b><br>2, 8, 18, 5         | 34<br><b>Se</b><br>2, 8, 18, 6         | 35<br><b>Br</b><br>2, 8, 18, 7         | 36<br><b>Kr</b><br>2, 8, 18, 8         |
| Gallium                                | Germanium                              | Arsenic                                | Selenium                               | Bromine                                | Krypton                                |
| 49<br><b>In</b><br>2, 8, 18, 18, 3     | 50<br><b>Sn</b><br>2, 8, 18, 18, 4     | 51<br><b>Sb</b><br>2, 8, 18, 18, 5     | 52<br><b>Te</b><br>2, 8, 18, 18, 6     | 53<br><b>I</b><br>2, 8, 18, 18, 7      | 54<br><b>Xe</b><br>2, 8, 18, 18, 8     |
| Indium                                 | Tin                                    | Antimony                               | Tellurium                              | Iodine                                 | Xenon                                  |
| 81<br><b>Tl</b><br>2, 8, 18, 32, 18, 3 | 82<br><b>Pb</b><br>2, 8, 18, 32, 18, 4 | 83<br><b>Bi</b><br>2, 8, 18, 32, 18, 5 | 84<br><b>Po</b><br>2, 8, 18, 32, 18, 6 | 85<br><b>At</b><br>2, 8, 18, 32, 18, 7 | 86<br><b>Rn</b><br>2, 8, 18, 32, 18, 8 |
| Thallium                               | Lead                                   | Bismuth                                | Polonium                               | Astatine                               | Radon                                  |

### Lanthanides

|                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |
|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| 57<br><b>La</b><br>2, 8, 18, 18, 9, 2 | 58<br><b>Ce</b><br>2, 8, 18, 20, 8, 2 | 59<br><b>Pr</b><br>2, 8, 18, 21, 8, 2 | 60<br><b>Nd</b><br>2, 8, 18, 22, 8, 2 | 61<br><b>Pm</b><br>2, 8, 18, 23, 8, 2 | 62<br><b>Sm</b><br>2, 8, 18, 24, 8, 2 | 63<br><b>Eu</b><br>2, 8, 18, 25, 8, 2 | 64<br><b>Gd</b><br>2, 8, 18, 25, 9, 2 | 65<br><b>Tb</b><br>2, 8, 18, 27, 8, 2 | 66<br><b>Dy</b><br>2, 8, 18, 28, 8, 2 | 67<br><b>Ho</b><br>2, 8, 18, 29, 8, 2 | 68<br><b>Er</b><br>2, 8, 18, 30, 8, 2 | 69<br><b>Tm</b><br>2, 8, 18, 31, 8, 2 | 70<br><b>Yb</b><br>2, 8, 18, 32, 8, 2 | 71<br><b>Lu</b><br>2, 8, 18, 32, 9, 2 |
| Lanthanum                             | Cerium                                | Praseodymium                          | Neodymium                             | Promethium                            | Samarium                              | Europium                              | Gadolinium                            | Terbium                               | Dysprosium                            | Holmium                               | Erbium                                | Thulium                               | Ytterbium                             | Lutetium                              |

### Actinides

|                                           |                                            |                                           |                                          |                                           |                                           |                                           |                                           |                                           |                                           |                                           |                                            |                                            |                                            |                                            |
|-------------------------------------------|--------------------------------------------|-------------------------------------------|------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| 89<br><b>Ac</b><br>2, 8, 18, 32, 18, 9, 2 | 90<br><b>Th</b><br>2, 8, 18, 32, 18, 10, 2 | 91<br><b>Pa</b><br>2, 8, 18, 32, 20, 9, 2 | 92<br><b>U</b><br>2, 8, 18, 32, 21, 9, 2 | 93<br><b>Np</b><br>2, 8, 18, 32, 22, 9, 2 | 94<br><b>Pu</b><br>2, 8, 18, 32, 24, 8, 2 | 95<br><b>Am</b><br>2, 8, 18, 32, 25, 9, 2 | 96<br><b>Cm</b><br>2, 8, 18, 32, 25, 9, 2 | 97<br><b>Bk</b><br>2, 8, 18, 32, 27, 8, 2 | 98<br><b>Cf</b><br>2, 8, 18, 32, 28, 8, 2 | 99<br><b>Es</b><br>2, 8, 18, 32, 29, 8, 2 | 100<br><b>Fm</b><br>2, 8, 18, 32, 30, 8, 2 | 101<br><b>Md</b><br>2, 8, 18, 32, 31, 8, 2 | 102<br><b>No</b><br>2, 8, 18, 32, 32, 8, 2 | 103<br><b>Lr</b><br>2, 8, 18, 32, 32, 9, 2 |
| Actinium                                  | Thorium                                    | Protactinium                              | Uranium                                  | Neptunium                                 | Plutonium                                 | Americium                                 | Curium                                    | Berkelium                                 | Californium                               | Einsteinium                               | Fermium                                    | Mendelevium                                | Nobelium                                   | Lawrencium                                 |